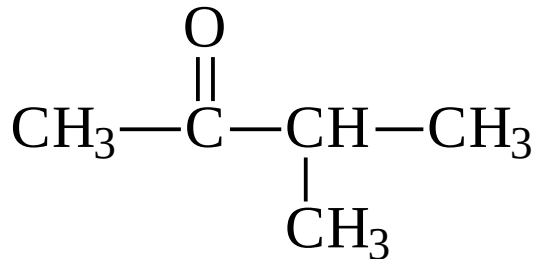
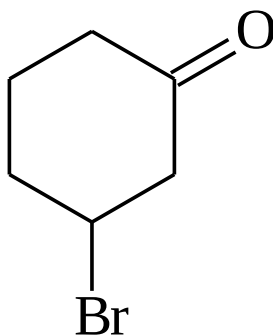


Aldehydes and Ketones

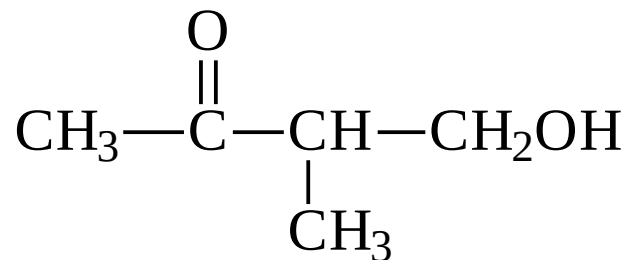
Examples



3-methyl-2-butanone



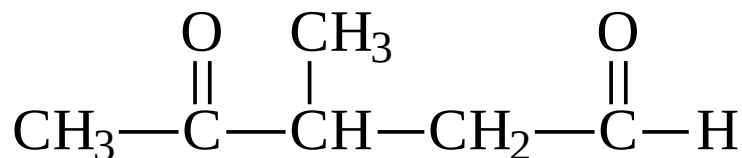
3-bromocyclohexanone



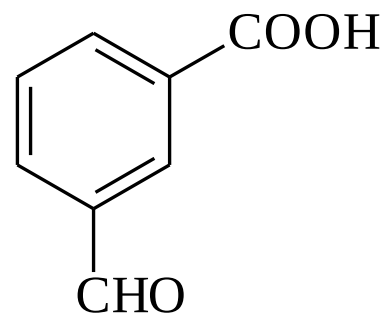
4-hydroxy-3-methyl-2-butanone

Name as Substituent

- On a molecule with a higher priority functional group, C=O is *oxo-* and -CHO is *formyl*.
- Aldehyde priority is higher than ketone.



3-methyl-4-oxopentanal



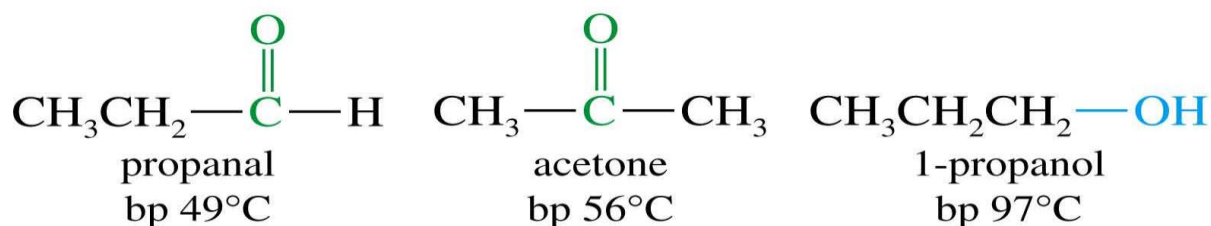
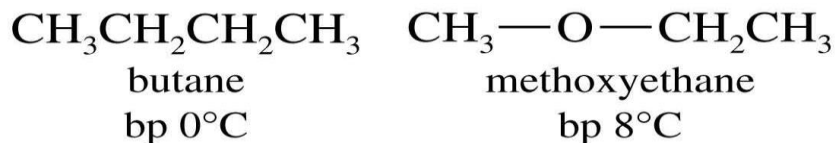
3-formylbenzoic acid 2

Aldehyde Common Names

- Use the common name of the acid.
- Drop *-ic acid* and add *-aldehyde*.
 - 1 C: formic acid, formaldehyde
 - 2 C's: acetic acid, acetaldehyde
 - 3 C's: propionic acid, propionaldehyde
 - 4 C's: butyric acid, butyraldehyde.

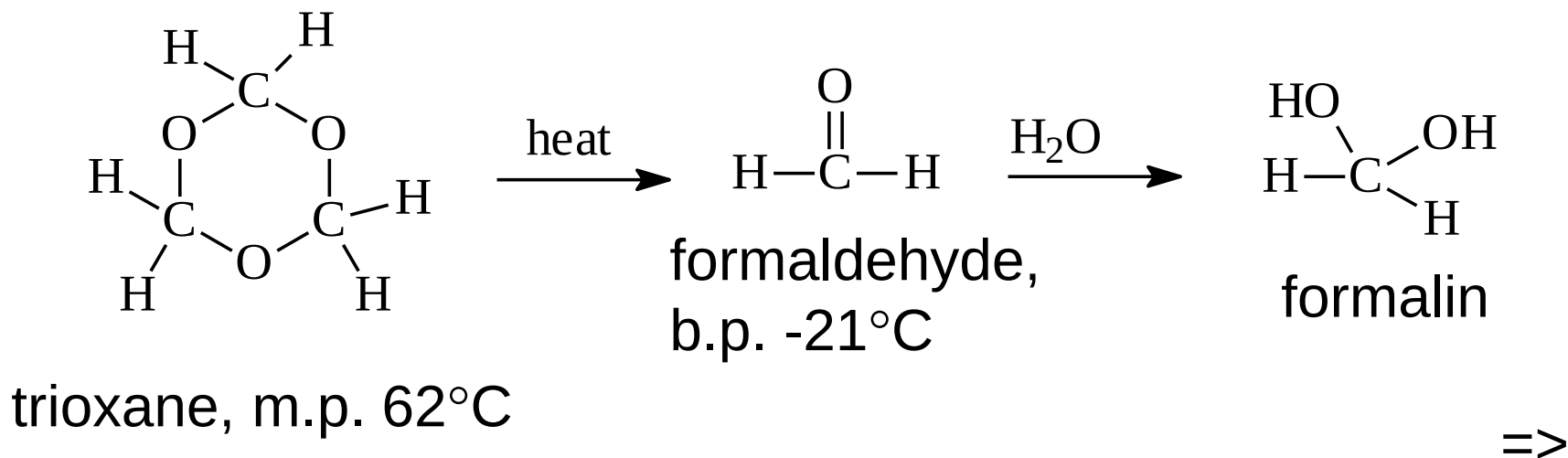
Boiling Points

- More polar, so higher boiling point than comparable alkane or ether.
- Cannot H-bond to each other, so lower boiling point than comparable alcohol.



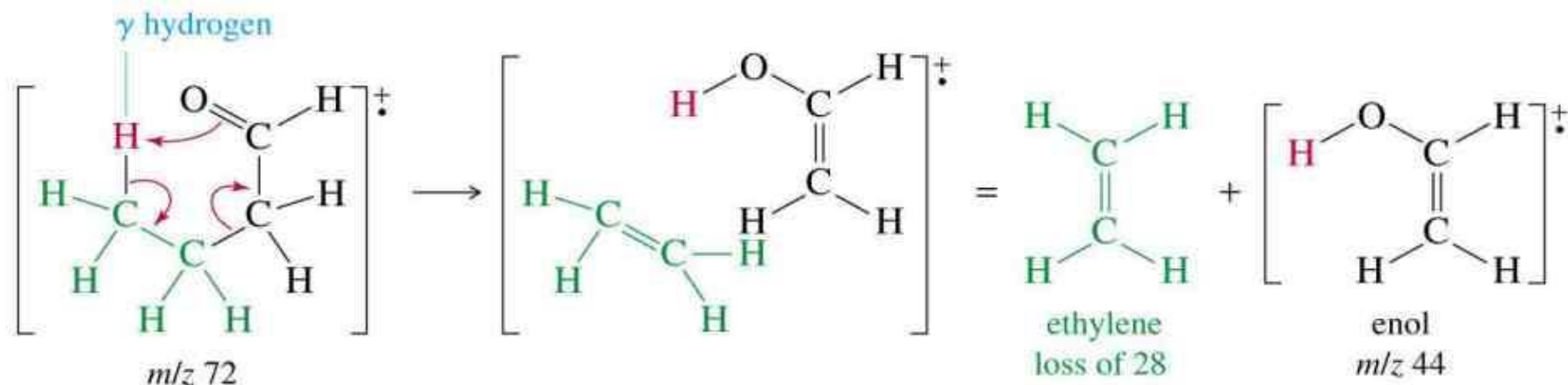
Formaldehyde

- Gas at room temperature.
- Formalin is a 40% aqueous solution.



McLafferty Rearrangement

- Loss of alkene (even mass number)
- Must have γ -hydrogen



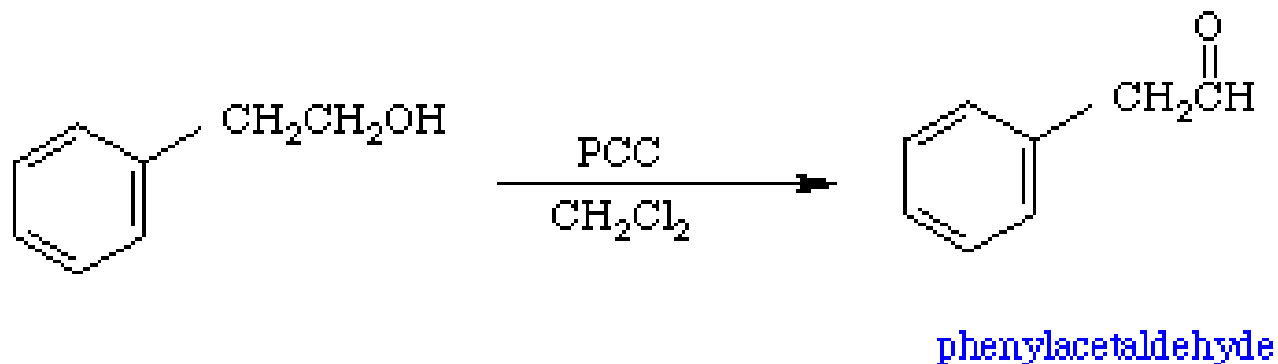
McLafferty rearrangement of butyraldehyde

=>

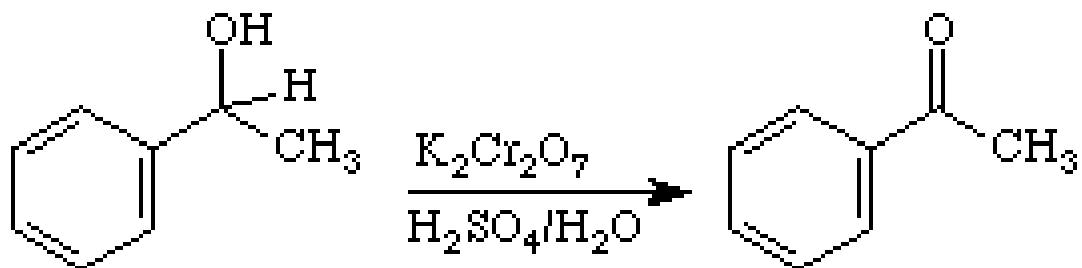
Synthesis

- Oxidation

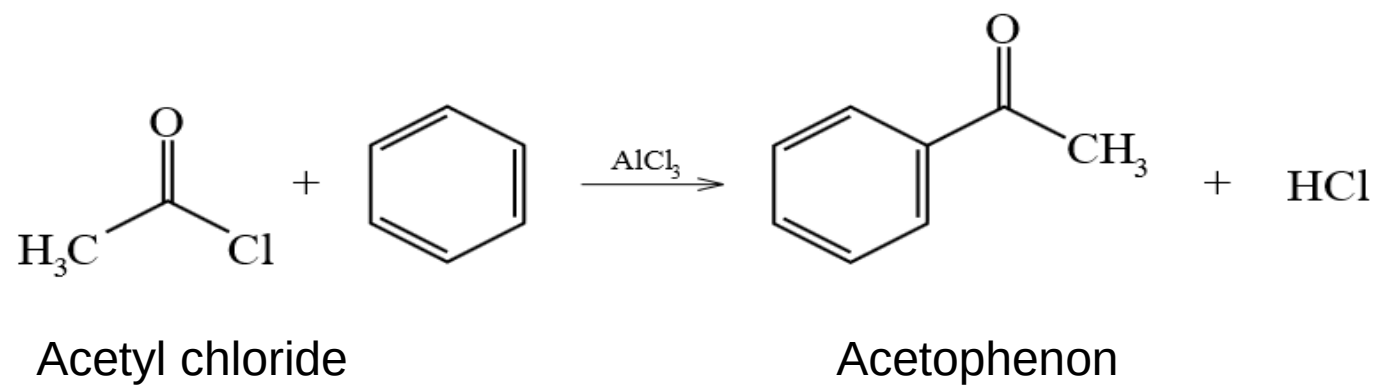
1° alcohol + PCC (Pyridinium chlorochromate) → aldehyde



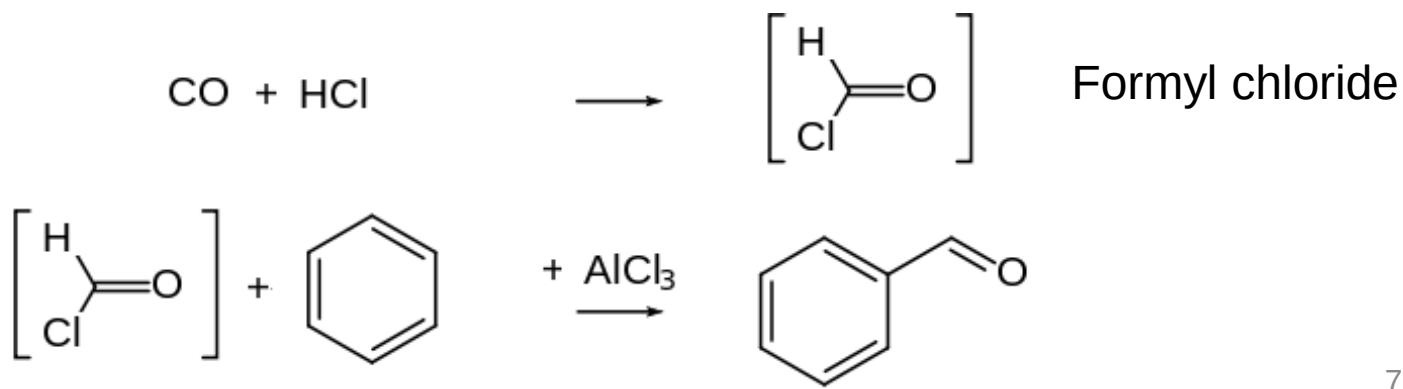
2° alcohol + $\text{Na}_2\text{Cr}_2\text{O}_7$ / $\text{K}_2\text{Cr}_2\text{O}_7$ → ketone



- Friedel-Crafts acylation
 Acid chloride/ AlCl_3 + benzene $\rightarrow$ ketone



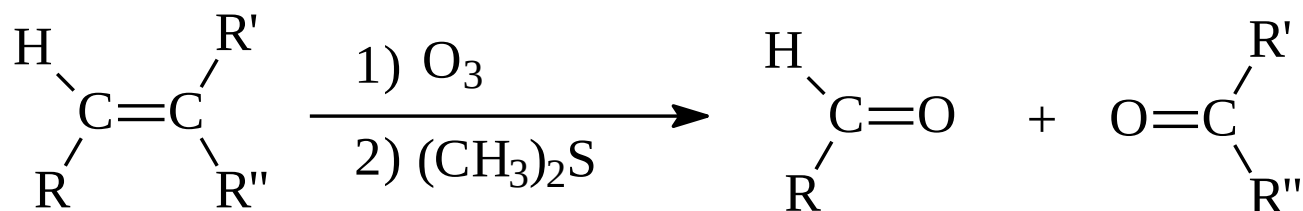
$\text{CO} + \text{HCl} + \text{AlCl}_3/\text{CuCl} + \text{benzene} \rightarrow \text{benzaldehyde}$ (Gatterman-Koch reaction)



Hydration of terminal alkyne

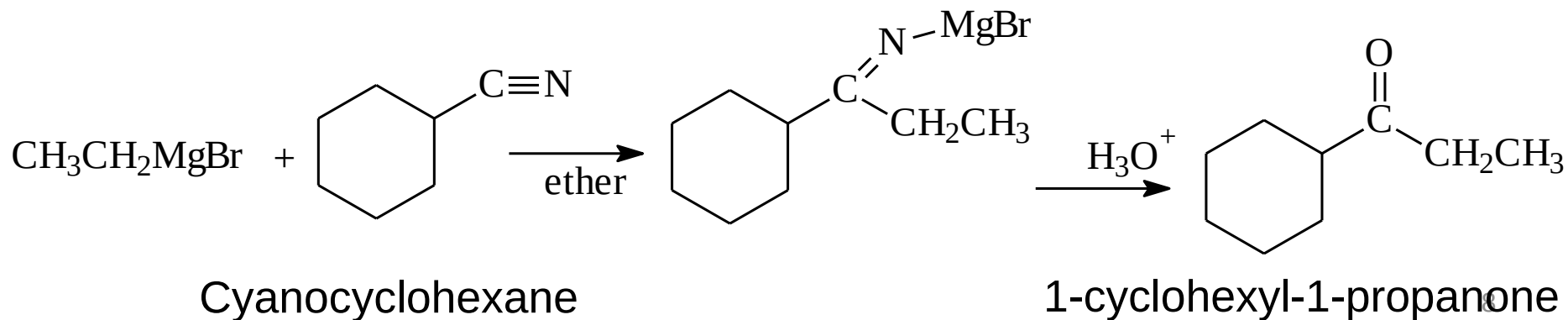
- Use HgSO_4 , H_2SO_4 , H_2O for methyl ketone
- Use Sia_2BH (Disiamyl borane) followed by H_2O_2 in NaOH for aldehyde.

Ozonolysis of alkenes.



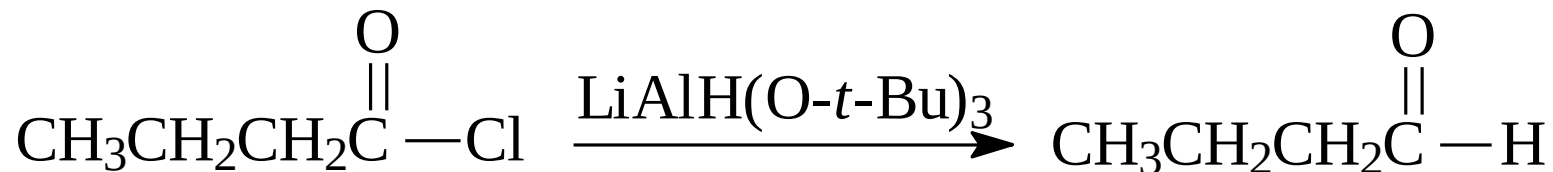
Ketones from Nitriles

- A Grignard or organolithium reagent attacks the nitrile carbon.
- The imine salt is then hydrolyzed to form a ketone.



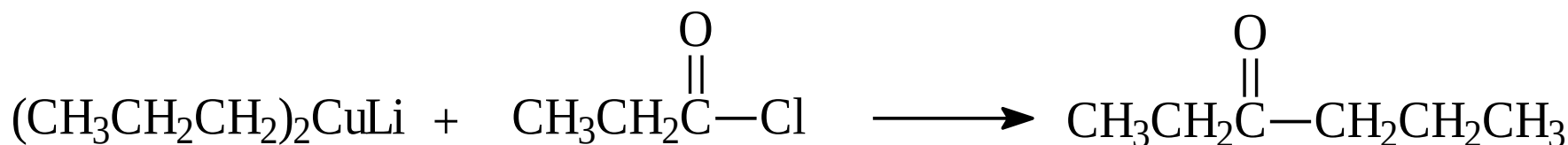
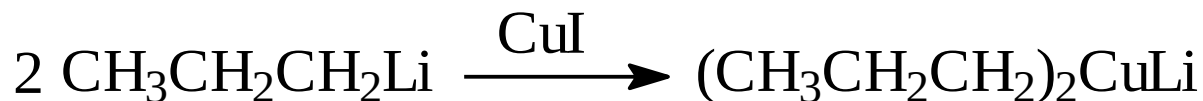
Aldehydes from Acid Chlorides

Use a mild reducing agent to prevent reduction to primary alcohol.



Ketones from Acid Chlorides

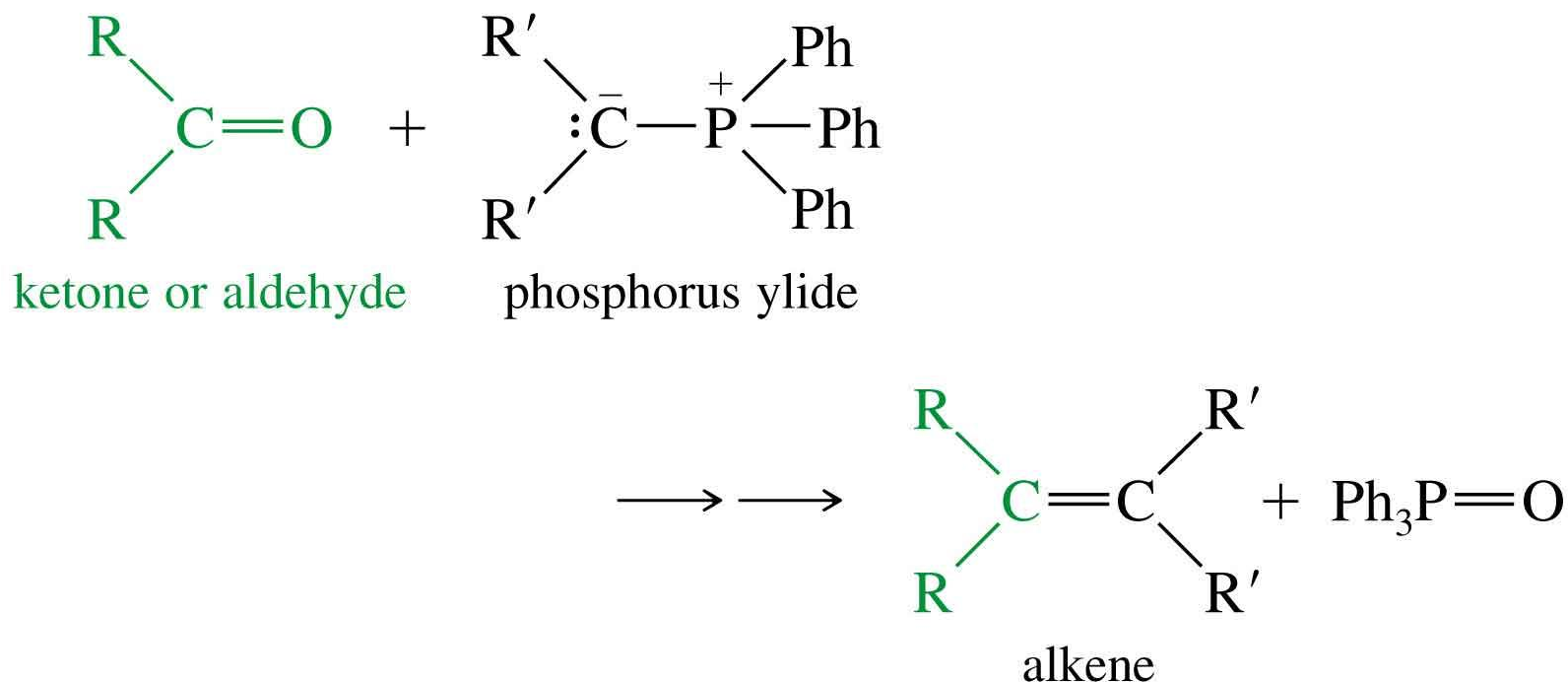
Use lithium dialkylcuprate (R_2CuLi), formed by the reaction of 2 moles of R-Li with cuprous iodide.



Wittig Reaction

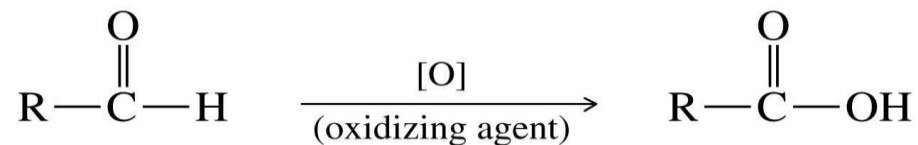
- Nucleophilic addition of phosphorus ylides.
- Product is alkene. C=O becomes C=C.

The Wittig reaction

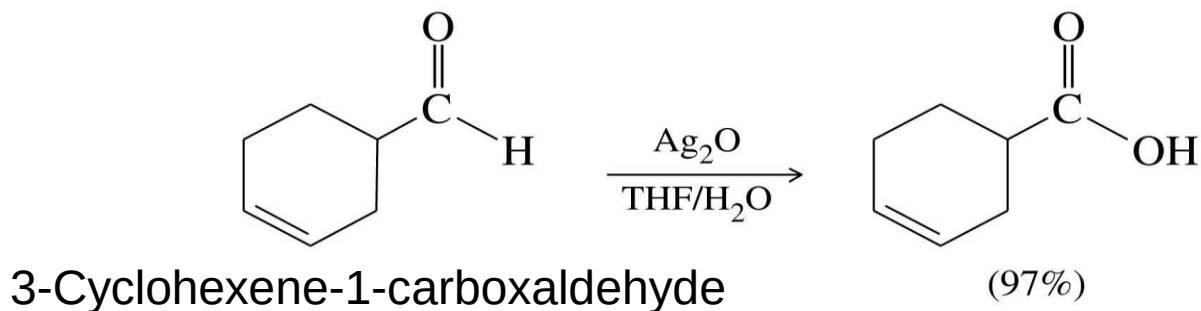
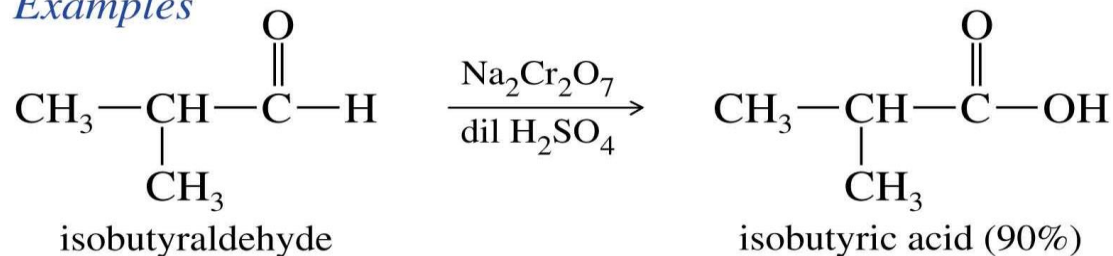


Oxidation of Aldehydes

Easily oxidized to carboxylic acids.

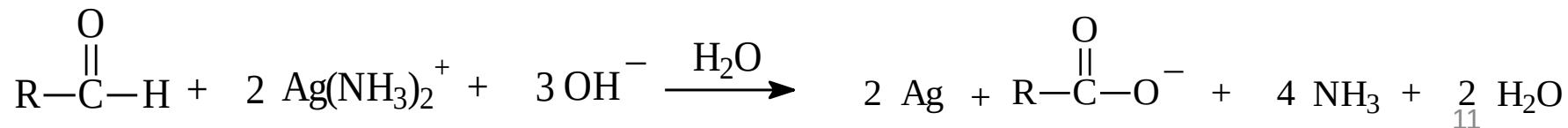


Examples



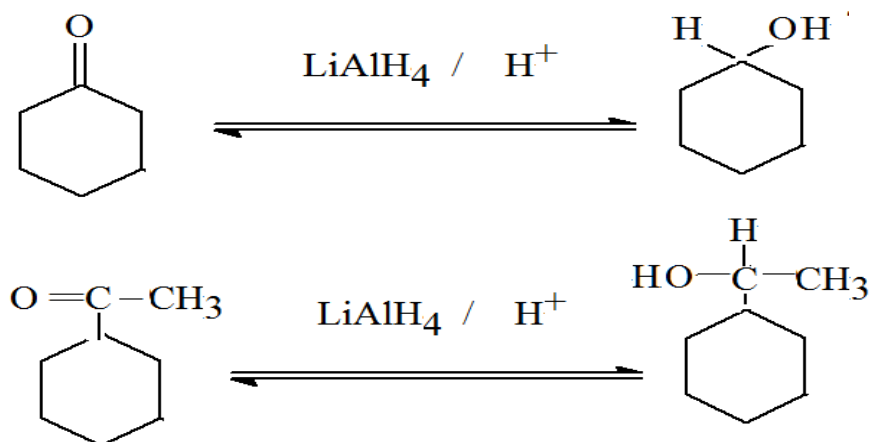
Tollens Test

- Aldehyde reaction forms a silver mirror.



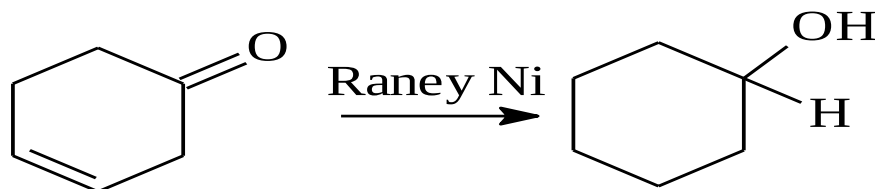
Reduction Reactions of Aldehydes & Ketones

- Sodium borohydride, NaBH_4 , reduces $\text{C}=\text{O}$, but not $\text{C}=\text{C}$.
- Lithium aluminum hydride, LiAlH_4 , much stronger, difficult to handle.
- Hydrogen gas with catalyst also reduces the $\text{C}=\text{C}$ bond.



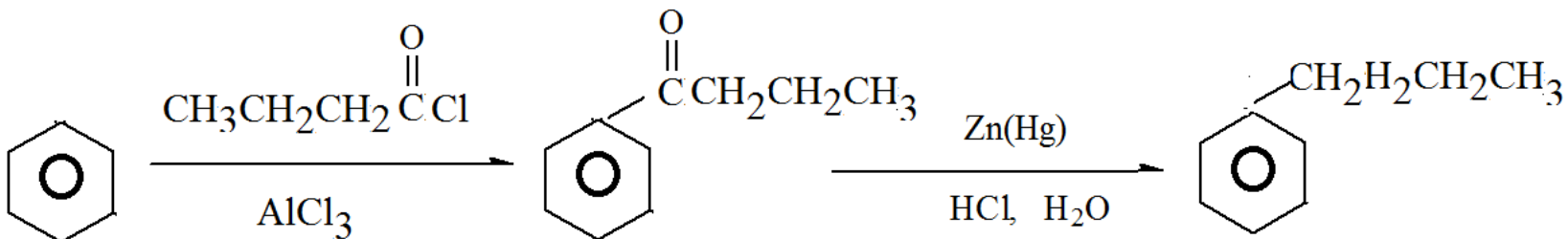
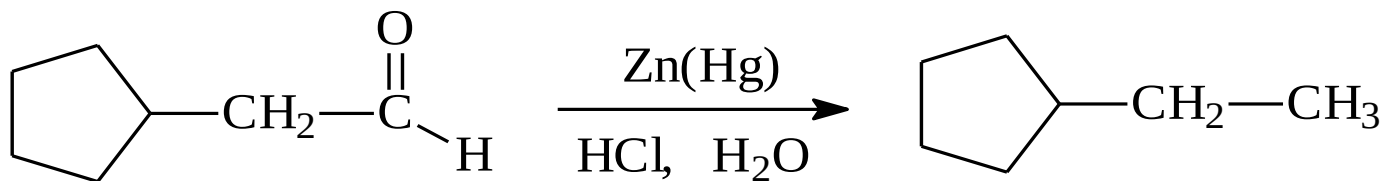
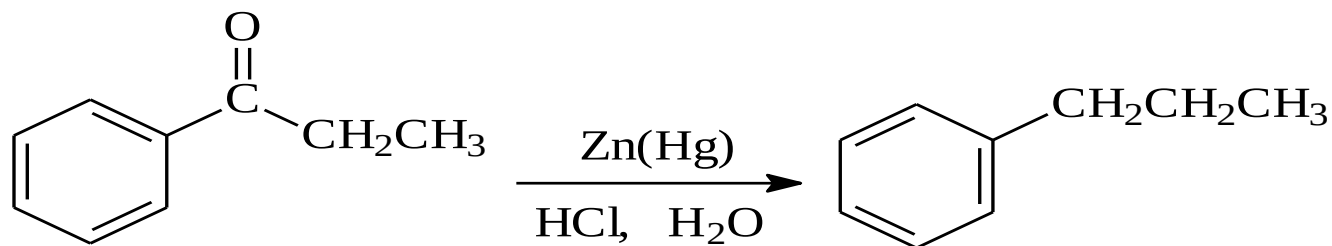
Catalytic Hydrogenation

- Widely used in industry.
- Raney nickel, finely divided Ni powder saturated with hydrogen gas.
- Pt and Rh also used as catalysts.



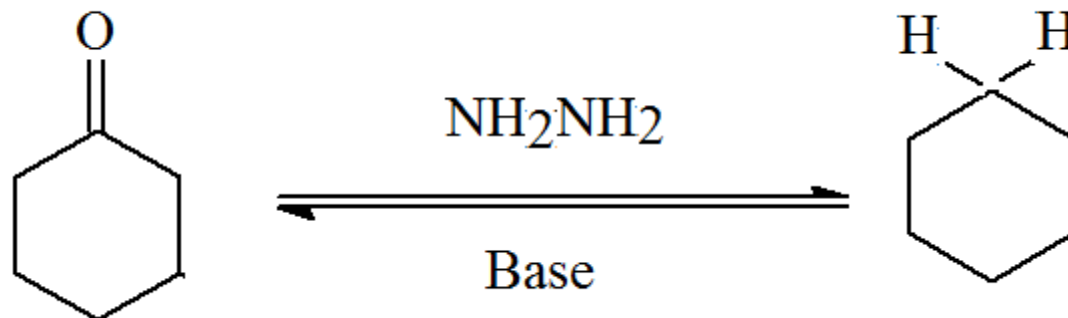
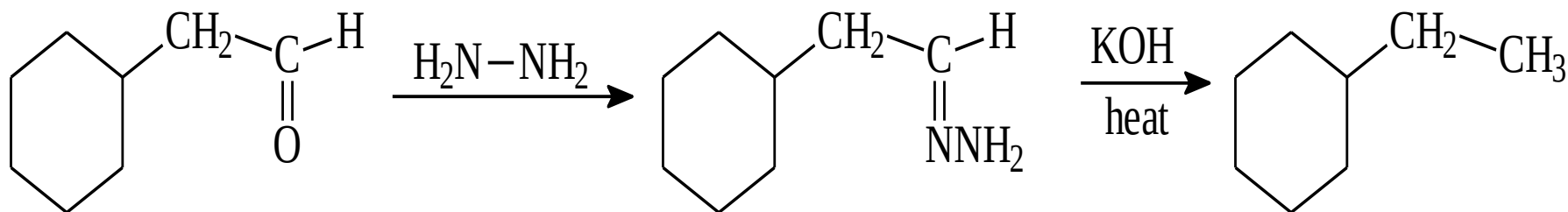
Reduction of Hydrocarbon -Deoxygenation

- Reduction of C=O to CH₂.
- Two methods:
 - Clemmensen reduction if molecule is stable in hot acid.



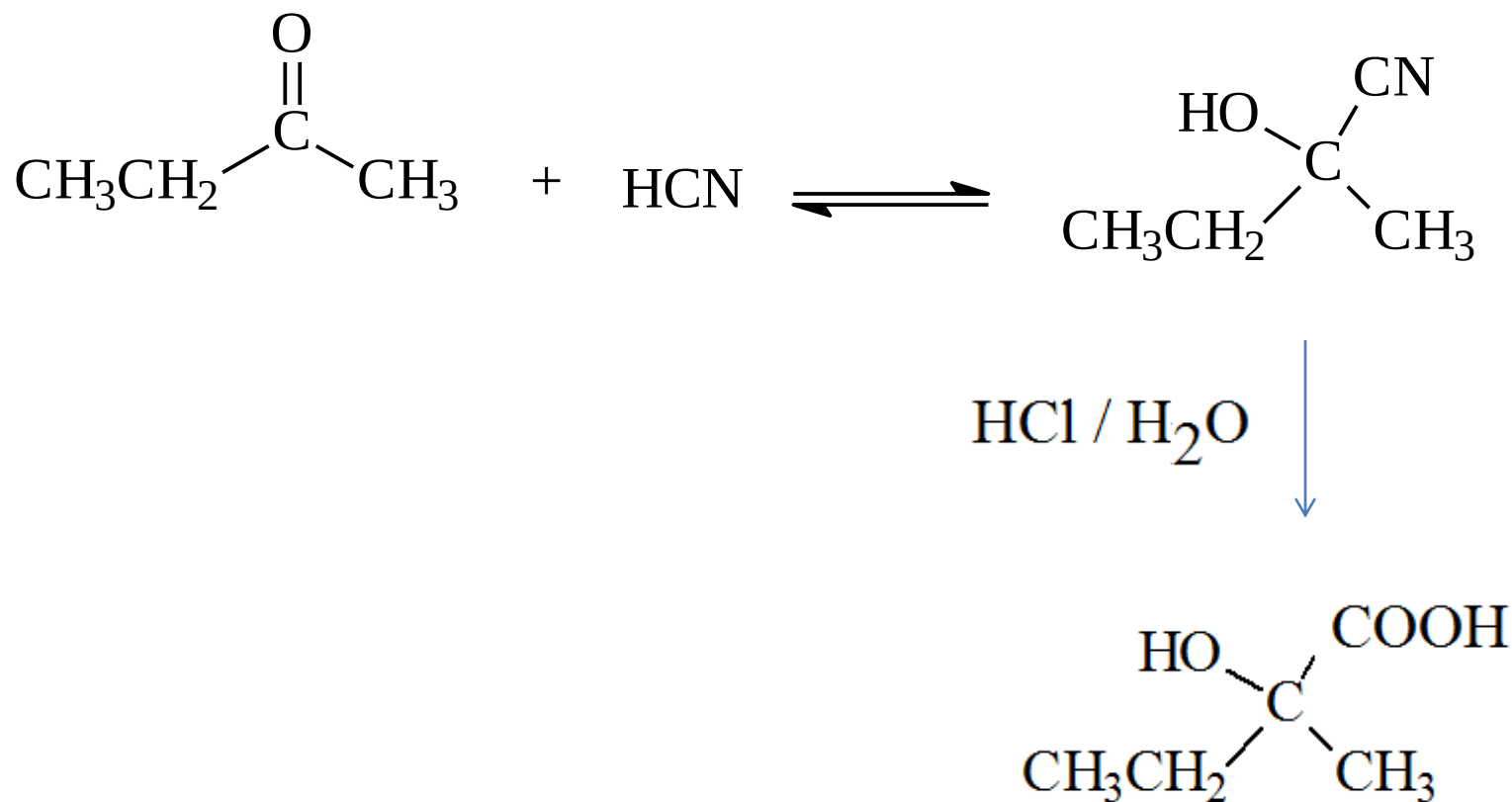
Wolff-Kishner reduction

if molecule is stable in very strong base. Form hydrazone, then heat with strong base like KOH or potassium *t*-butoxide.

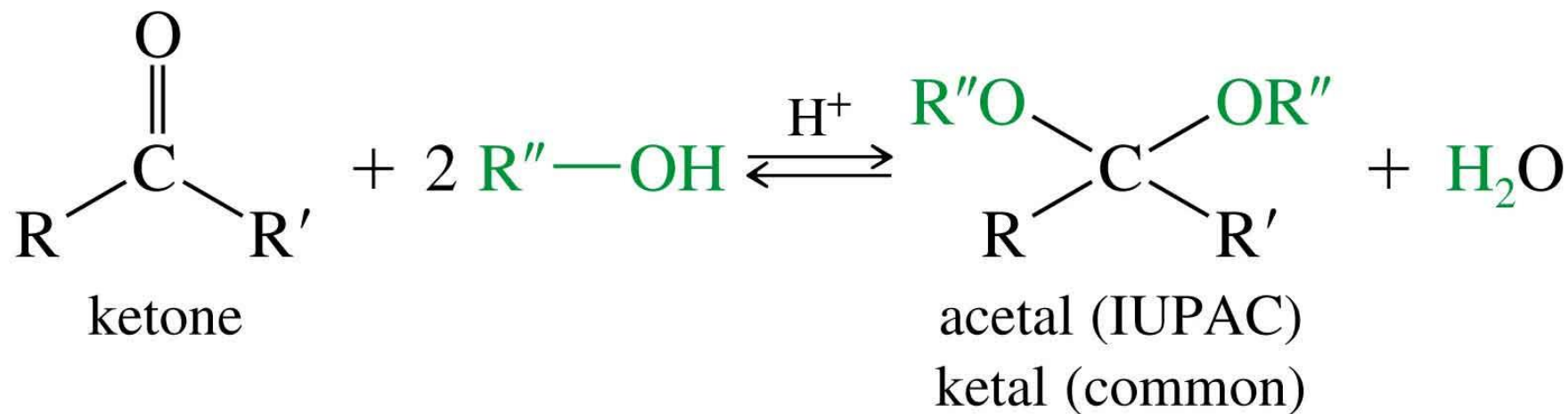
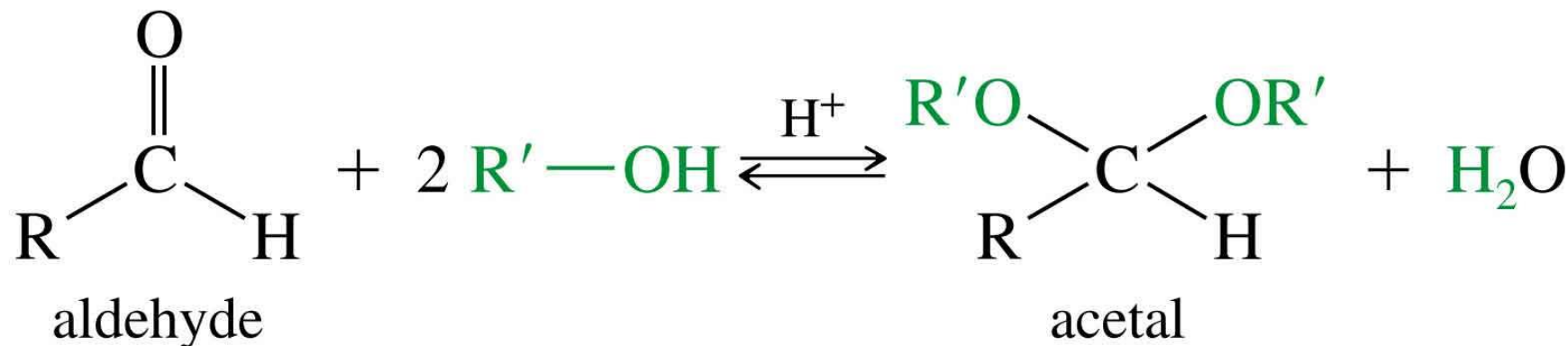


Addition of Cyanide; Cyano – hydrin formation

- Use NaCN, HCN or KCN in base to add cyanide, then protonate to add H.
- Reactivity formaldehyde > aldehydes > ketones >> bulky ketones.



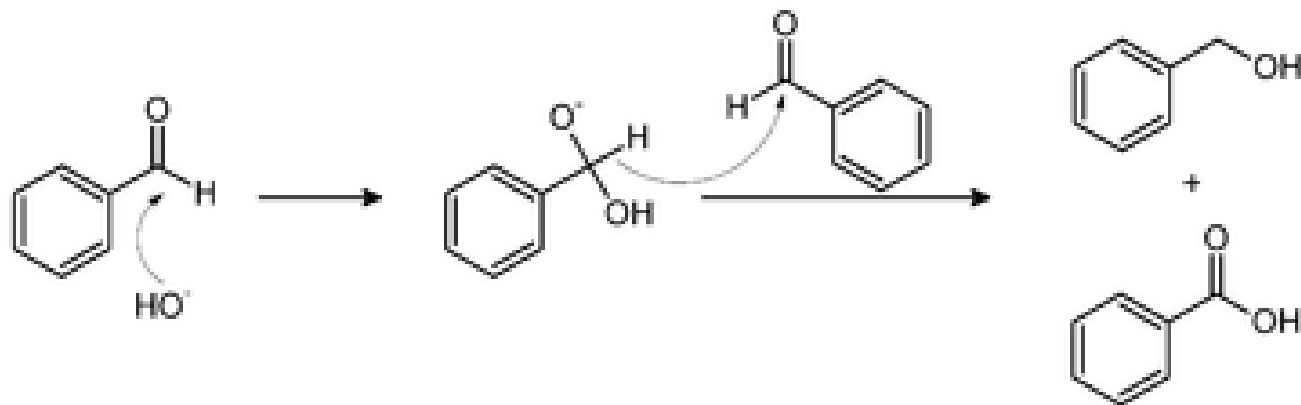
Addition of Alcohols



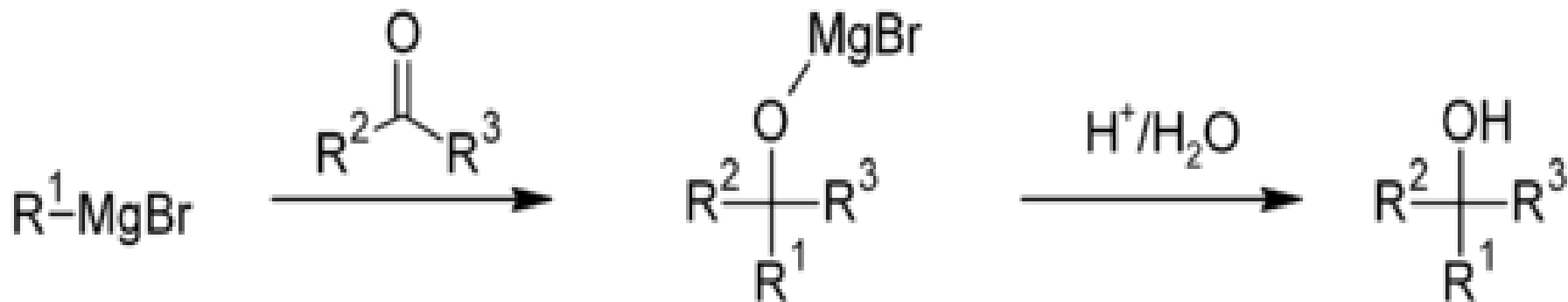
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Cannizzaro reaction

2 molecules of aldehydes converts to a carboxylic acid (salt) and an alcohol

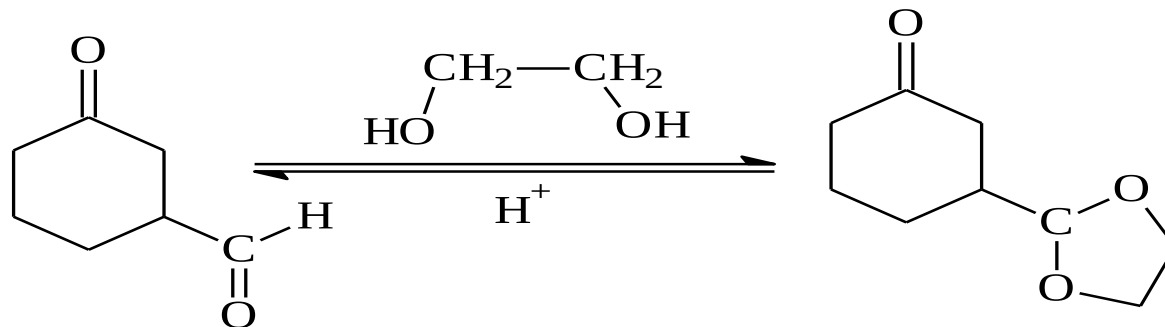


Grignard reaction



Selective Reaction of Ketone

- Hydrolyze easily in acid, stable in base.
- Aldehydes more reactive than ketones.



- React with strong nucleophile (base)
- Remove protective group.

